Benjamin Purinton, M.Sc.

Ph.D. Student – Remote Sensing, University of Potsdam – Dual Citizen of USA & Germany
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Education

Ph.D. Student, Remote Sensing, University of Potsdam, Germany ..... Since February 2017 (Expected December 2019)

Master of Science, Geology, University of Potsdam, Germany ..... November 2016

Bachelor of Arts, Earth and Environmental Sciences, Wesleyan University, USA ..... May 2013

Technical Proficiencies

Software:

- Proficient on Windows, Mac, and Linux operating systems
- GIS – GDAL/OGR, QGIS, ArcGIS
- Computer Coding – Python, MATLAB, Command-Line and Bash Scripting
- Remote Sensing – ENVI, PCI-Geomatica, SNAP
- Point Clouds – Agisoft Metashape, CloudCompare, LAStools, PDAL
- Other – Adobe Illustrator, Microsoft Office Suite, RTKLIB GNSS Correction

Skills:

- Optical and radar satellite data management and processing including, but not limited to: interferometry, georeferencing, orthorectification, photogrammetry, and landscape feature extraction
- Python programming including the development of image processing software for grain-size analysis
- Collection and processing of differential GPS (dGPS) measurements on various units
- DEKRA certified drone pilot with license to operate in Germany
- Structure-from-Motion with Multi-View Stereo (SfM-MVS) processing of photo surveys from drones and hand-held cameras to generate dense point clouds and digital surface models

Honors and Awards

Peirce Prize for Excellence in Geology, Wesleyan University, USA ..... May 2013

Phi Beta Kappa Distinction, Wesleyan University, USA ..... May 2013

Senior Thesis High Honors, Wesleyan University, USA ..... May 2013

Academic Activities

Geologic Field Mapping Course of Miami University (Ohio), Wyoming, USA .....June – July 2012

Research Intern, Columbia University's Lamont-Doherty Earth Observatory, New York, USA .....June – August 2011

Invited Talks

German Research Centre for Geosciences (GFZ), Section 5.1: Geomorphology ..... October 2018

Teaching

Introduction to GIS, University of Potsdam .....Block Course February 2017 and 2019

Terrestrial and Airborne Lidar and Photogrammetry Systems, University of Potsdam .....Summer Semester 2019

Languages

English – Native
German – Intermediate Level (CEFRL Level B2 Equivalent)

Peer Reviewed Journal Articles

Purinton, B. and Bookhagen, B.: Introducing PebbleCounts: A semi-automated grain-sizing tool for photo surveys of dynamic gravel-bed rivers, Earth Surface Dynamics Discussion, https://doi.org/10.5194/esurf-2019-20, *in review*, 2019.

Purinton, B. and Bookhagen, B.: Measuring decadal vertical land-level changes from SRTM-C (2000) and TanDEM-X (~ 2015) in the south-central Andes, Earth Surface Dynamics, 6, 971-987, https://doi.org/10.5194/esurf-6-971-2018, 2018.

Purinton, B. and Bookhagen, B.: Validation of digital elevation models (DEMs) and comparison of geomorphic metrics on the southern Central Andean Plateau, Earth Surface Dynamics, 5, 211-237, https://doi.org/10.5194/esurf-5-211-2017, 2017.

Selected Conferences

Purinton, B. and Bookhagen, B.: Decadal rates and spatial distribution of gravel-bed river aggradation and incision from spaceborne DEMs: Applications in the Northwest Argentine Andes, EGU General Assembly, Vienna, Austria, 2018.

Purinton, B. and Bookhagen, B.: Elevation validation and geomorphic metric comparison with focus on ASTER GDEM2, SRTM-C, ALOS World 3D, and TanDEM-X, EGU General Assembly, Vienna, Austria, 2017.

Purinton, B. and Bookhagen, B.: Quality assessment and geomorphic application of digital elevation models in the central Andes with focus on TSX/TDX, TerraSAR-X / TanDEM-X Science Meeting, Oberpfaffenhofen, Germany, 2016.

Purinton, B.: Hydrologic and geomorphic impacts of the 2010 Fourmile Canyon Fire, Boulder County, CO, GSA Northeastern Section - 48th Annual Meeting, New Hampshire, USA, 2013.